



“ A PROJECTILE WOULD BE **MOVED BY AN IMPETUS...** BY THE THROWER AND WOULD **CONTINUE TO BE MOVED AS LONG AS THE IMPETUS REMAINED STRONGER THAN THE RESISTANCE.** ”

Jean Buridan, French priest, in *Questions on the Physics of Aristotle*, c.1357

A catapult in the fortress of Edessa hurls a ball at a siege tower. According to Jean Buridan's theory, the catapult has imparted impetus to the projectile.

IN THE LATE 1340S, A NEW AND TERRIBLE DISEASE STRUCK

Europe, the Middle East, and North Africa. The Black Death was an **epidemic of bubonic plague** that infected humans via rat fleas. It was discovered later that the cause was a bacterium called *Yersinia pestis*. The plague

Plague victims

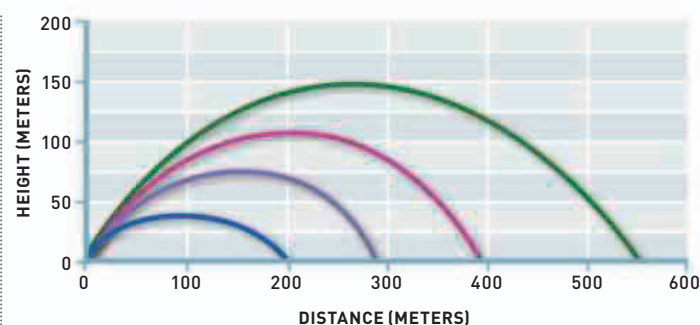
This 15th-century Swiss manuscript shows victims of the plague with the characteristic swellings, or buboes, covering large parts of their bodies.

reached Constantinople in 1347 and spread by ship throughout the Mediterranean, arriving in France and England in 1348. The disease began with swellings, or buboes, in the groin and armpit, followed by the spread of black spots over the body and high fever. It caused millions of deaths across Europe.

Contemporary physicians, who **had no cure for the plague**, believed that it was caused by putrefaction in the air brought on

by humidity or rotting corpses. Remedies included controlling the body's heat by avoiding "putrefying" foods, such as meat and fish, fumigating rooms, and wearing pomanders infused with spices close to the nose.

Although doctors had failed to control the plague, after its end a renewed energy was given to medical science. By 1351, Padova had 12 medical professors (as against three in 1349). **Measures to promote public health** were



also enacted. In 1377, the Republic of Ragusa (Dubrovnik) ordered a **quarantine** of 30 days for anyone coming from plague-infected areas, as did Marseilles in 1383. By 1450 Milan would establish a **permanent board of health**, and **health passports** were introduced in Italy in 1480.

Aristotle's explanation of projectiles in motion had long puzzled scholars. In 1357, the French priest **Jean Buridan** (c.1300–58) published *Questions on the Physics of Aristotle*. He pointed out that a thrown stone continues to move even out of

Free fall

The *impetus* provided by throwing an object to a greater height means it travels farther before downward forces pulls it back to the ground.

contact with the thrower. He theorized that the person throwing an object imparts a force to it, which he called **impetus**, and this causes it to continue to move, so long as the resistance of the air does not stop it. He believed the amount of impetus in an object depended on the amount of matter in it, so that feathers would not move quickly when thrown, whereas heavier objects would.



50 PERCENT
THE **PROBABLE DEATH RATE**
IN EUROPE DURING THE PEAK
OF **THE BLACK DEATH** IN THE
MID-14TH CENTURY

1351 Black Death epidemic peaks out. At its peak, it had killed about 50 percent of Europe's population

1357 French priest Jean Buridan publishes his theory of *impetus*